

34721/34722 Linear Control Design 1

Spring Semester

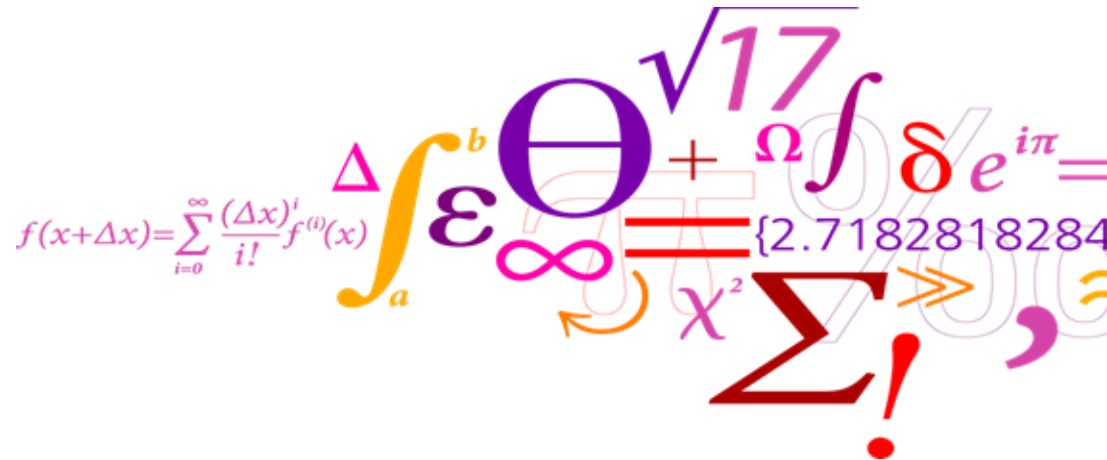
Silvia Tolu - Dimitrios Papageorgiou



Lecture 9/13

Schedule (13:00-15:00)

- **Closed-loop system time-domain specifications**
- **Bandwidth**
- **Control design based on specifications**



What you will learn today

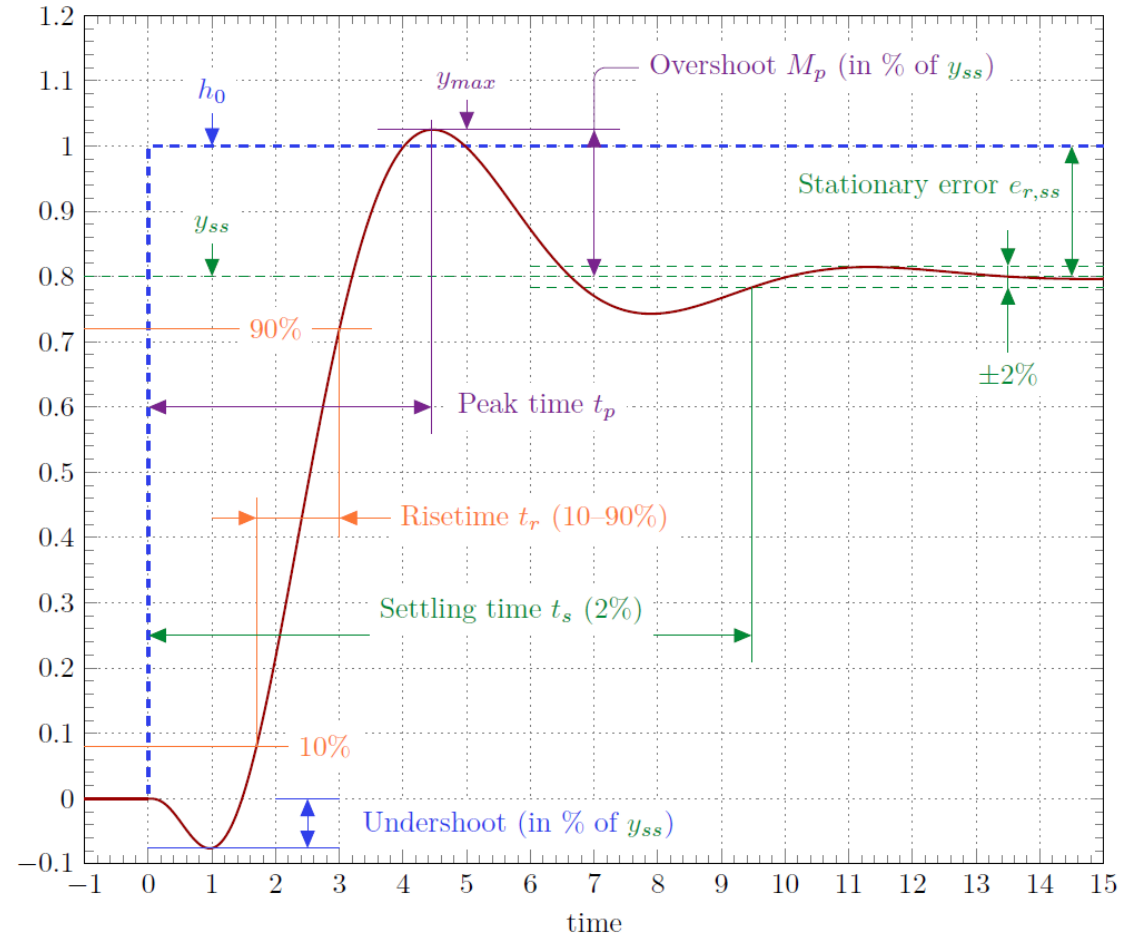
After this lecture you will be able to

- Transform closed-loop specification into open-loop specifications for design.
- Decide which controller parts are needed for a given system.
- Design feedback controllers based on closed-loop system specifications.
- Interpret Bode plots for control design

Closed-loop specifications

Consider the output response of a system to a **step** input of size h_0 .

- **Rise time (t_r)** : Time from 10% to 90% of final value.
- **Settling time (t_s)** : Time until the output is contained in $\pm 2\%$ of final value.
- **Peak time (t_p)** : Time until maximum value.



Closed-loop specifications

Consider the output response of a system to a **step** input of size h_0 .

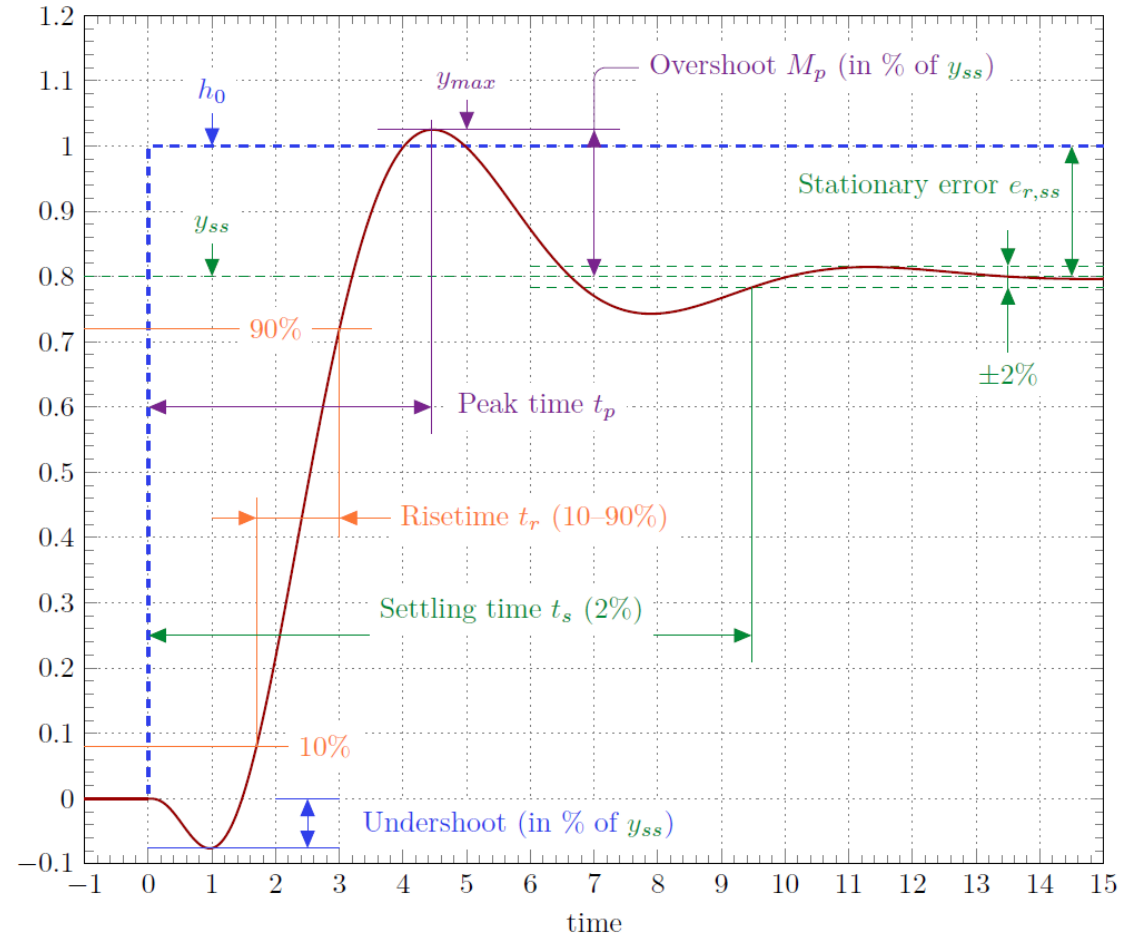
- **Overshoot (M_p)** : Difference between maximum and steady state value (often in %).

$$M_p = \frac{y_{max} - y_{ss}}{y_{ss}}$$

- **Stationary (static, steady-state) error (e_{ss})** : Difference between step reference and steady state value (sometimes in %).

$$e_{ss} = |h_0 - y_{ss}| \text{ or } e_{ss} = \left| \frac{h_0 - y_{ss}}{y_{ss}} \right| 100\%$$

- **(Undershoot** : Minimum minus initial value).



Closed-loop specifications

Closed-loop specifications

- Informative & intuitive
- It's how actual specifications are described in industry

- Rise time (t_r)
- Settling time (t_s)
- Peak time (t_p)
- Overshoot (M_p)
- Stationary (static, steady-state) error (e_{ss})

...

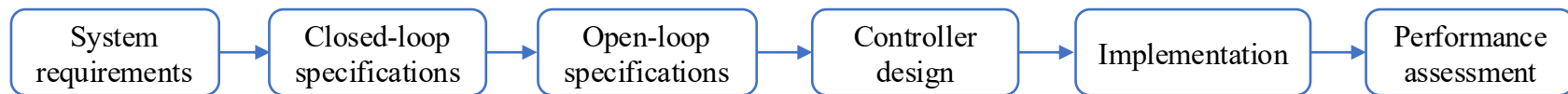


Open-loop specifications

- Can be used for control design
- Can be checked/verified in the design phase (before implementation)

- Phase margin (γ_M)
- Gain margin (K_M)
- Controller type (P,PI,PI-Lead...)
- Controller gains (K_p)
- Crossover frequency (ω_c)

...



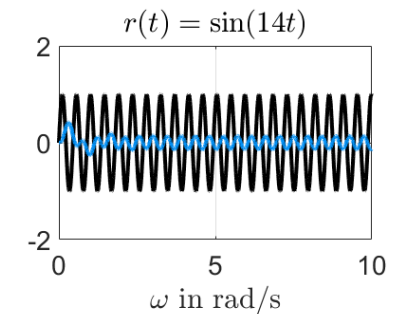
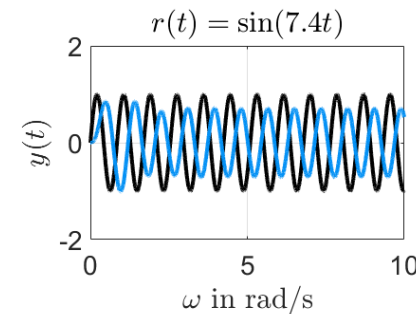
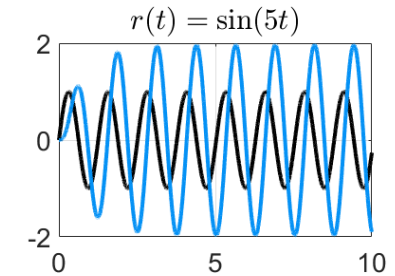
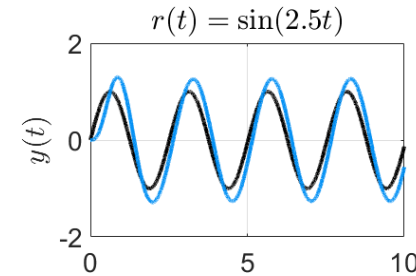
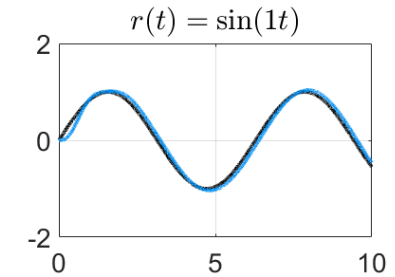
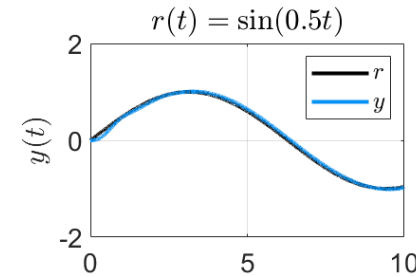
Bandwidth

Ideally, a closed-loop system should behave like a 1st or a 2nd order lowpass filter with

- Unit static gain
- Low (or no) overshoot
- Short rise time
- Short settling time
- Large **bandwidth**

Bandwidth: The frequency at which the amplitude drops by 3 dB compared to the low frequency horizontal asymptote.

Up to which frequency the system substantially responds to the input

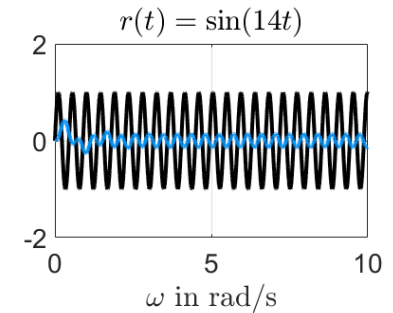
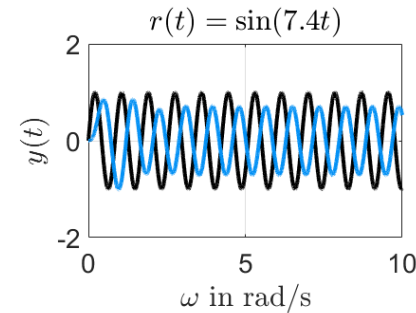
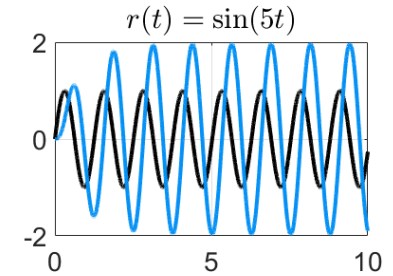
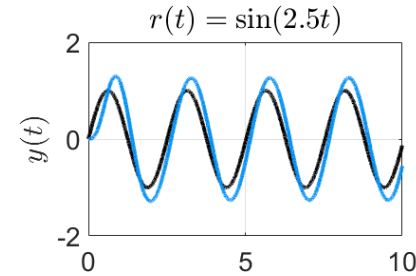
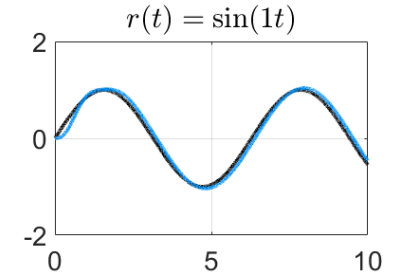
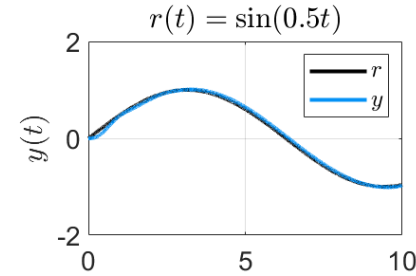
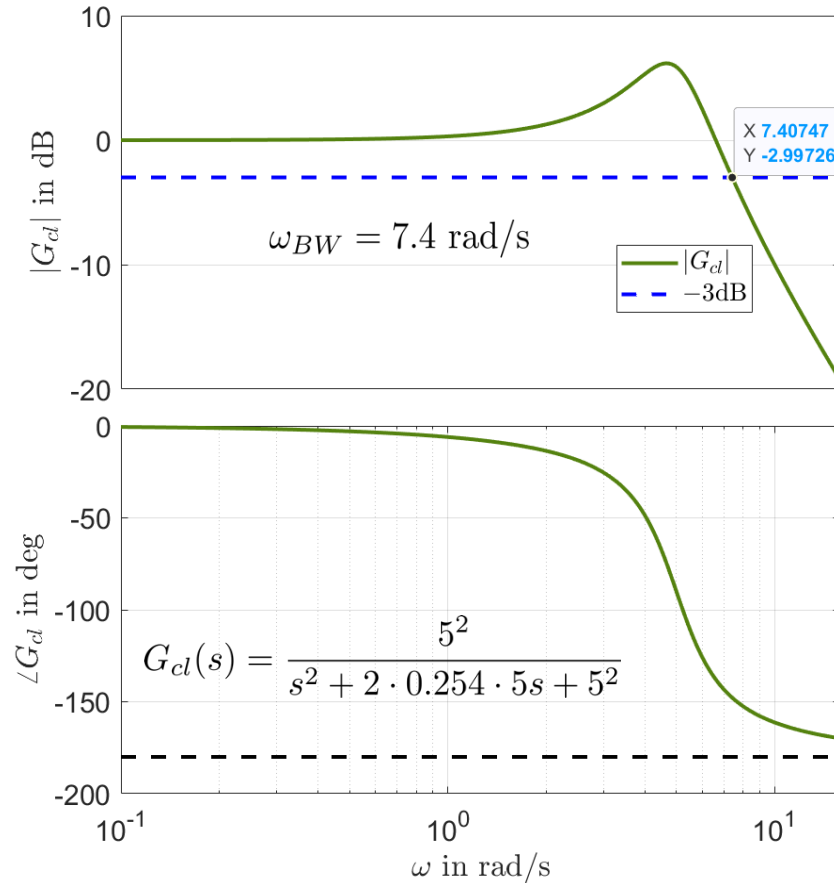


Bandwidth

stable_spring_simulator.m

Bandwidth_bode_sine.m

Closed-loop Bode plot

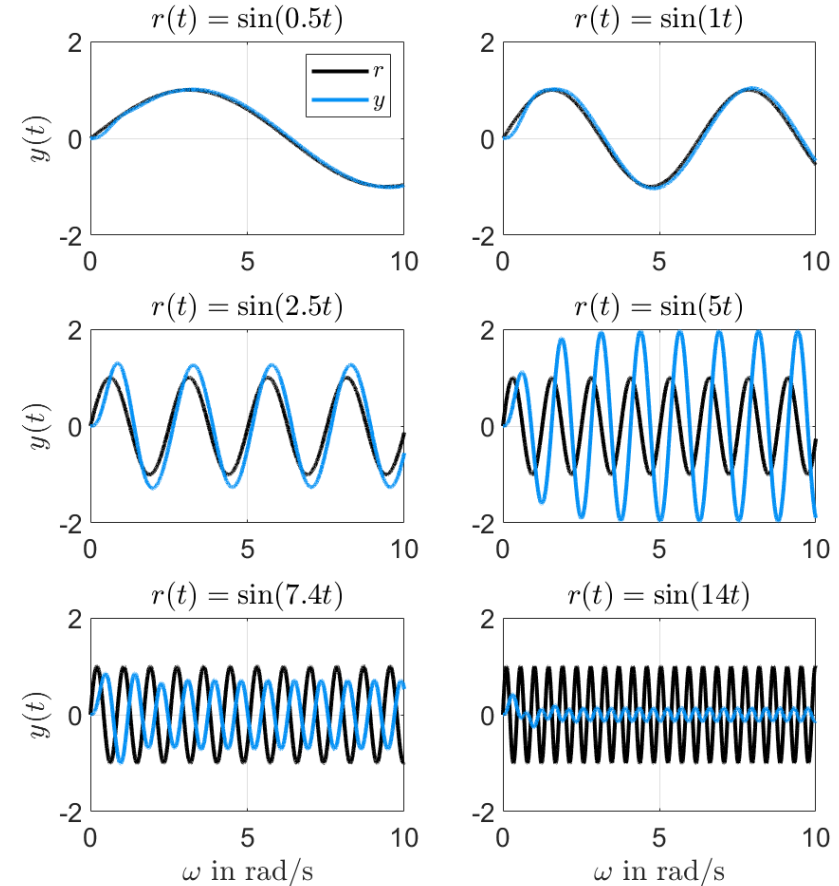


Bandwidth

Bandwidth: The frequency at which the amplitude drops by 3 dB compared to the low frequency horizontal asymptote.

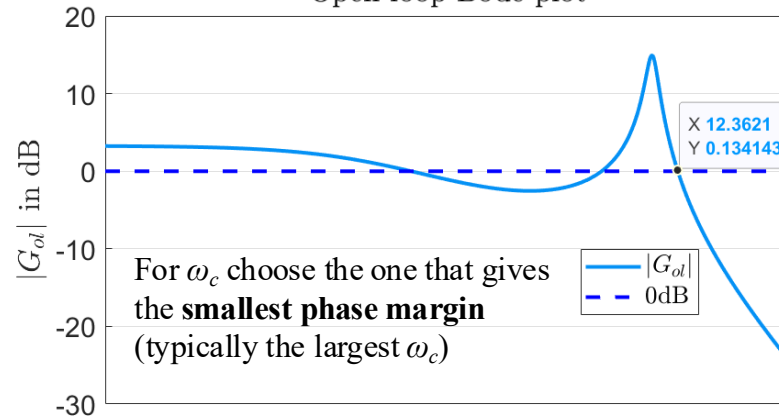
- Higher bandwidth $\rightarrow$ the system can respond to faster inputs (more aggressive)
- It is a property of the **closed-loop** system
- It is meaningful for stable closed-loop systems (with finite low frequency amplitude)

$$|G_{cl}(j\omega_{BW})| = \underbrace{\frac{\sqrt{2}}{2}}_{-3 \text{ dB}} |G_{cl}(0)|$$

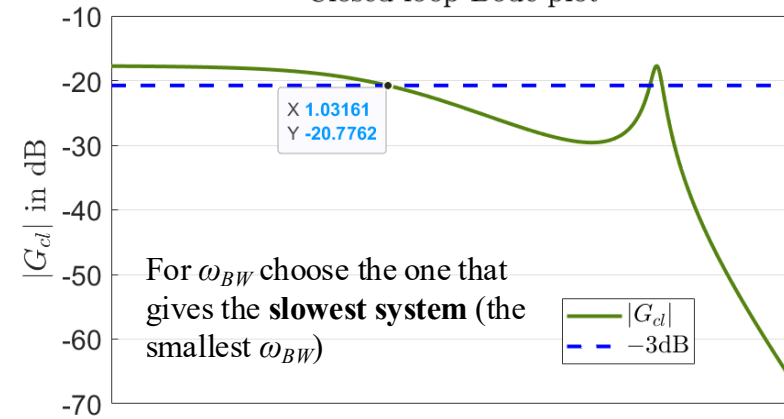


Bandwidth

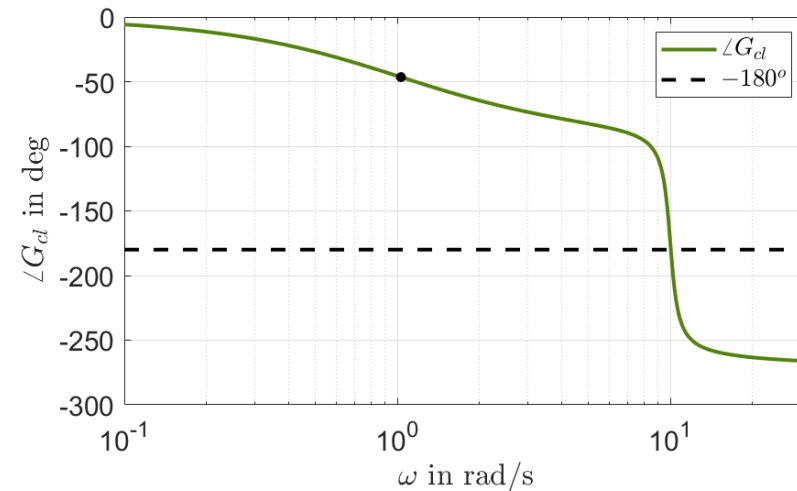
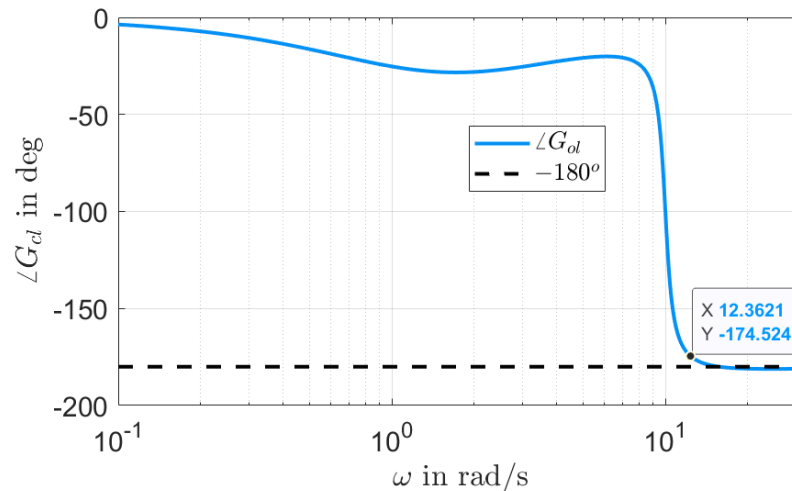
Open-loop Bode plot



Closed-loop Bode plot



What happens if we have multiple 0 dB or -3 dB frequencies? → choose the most limiting of those.



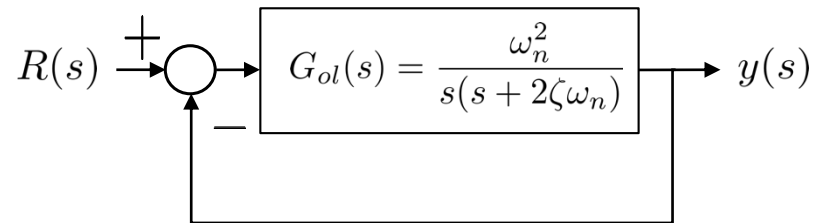
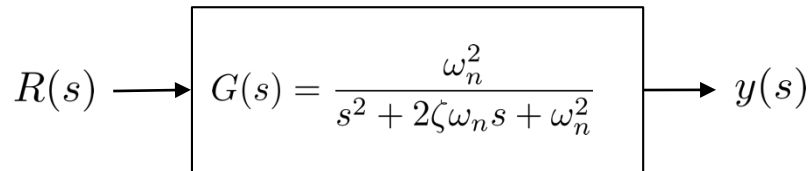
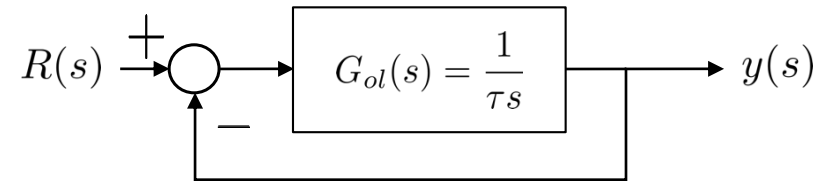
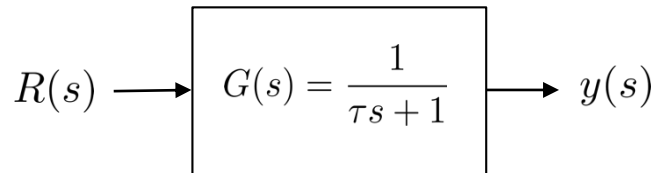
Control design based on specifications

Ideally, a closed-loop system should behave like a 1st or a 2nd order lowpass filter with

- Unit static gain
- Low (or no) overshoot
- Short rise time
- Short settling time
- Large **bandwidth**

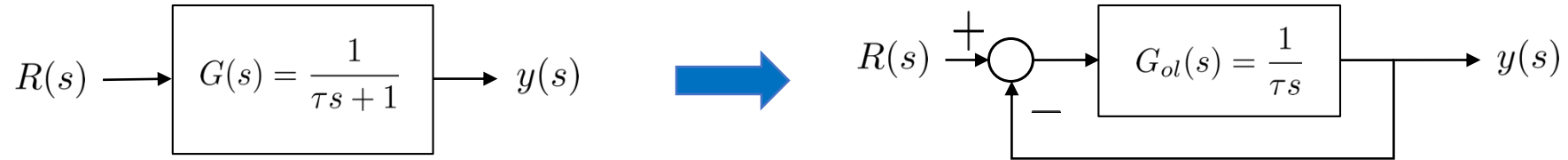


- Phase margin (γ_M)
- Controller gains (K_P)
- Crossover frequency (ω_c)



Control design based on specifications

If we want a closed-loop system with a 1st order lowpass filter behaviour of time constant τ , then



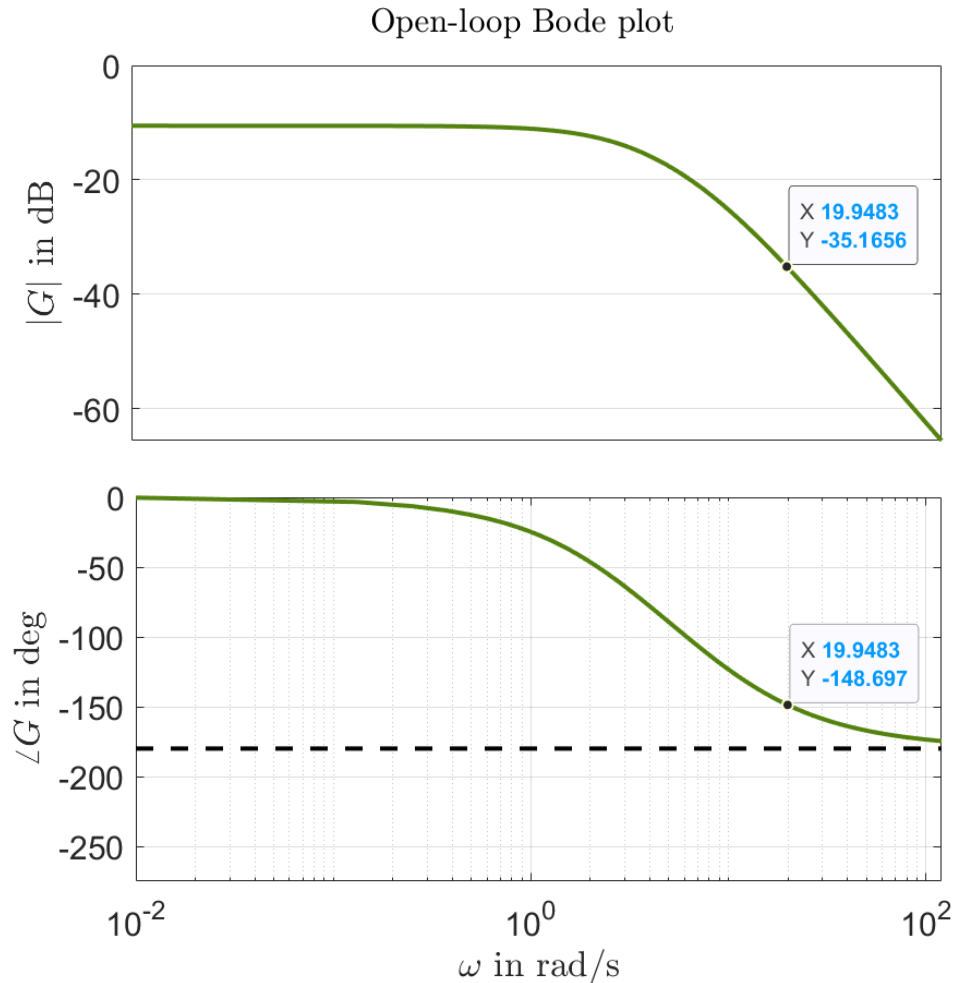
- Crossover frequency: $|G_{ol}(j\omega_c)| = 1 \Leftrightarrow \frac{1}{\omega_c \tau} = 1 \Leftrightarrow \boxed{\omega_c = \frac{1}{\tau}}$
- Bandwidth: $|G_{cl}(j\omega_{BW})| = \underbrace{\frac{\sqrt{2}}{2}}_{-3 \text{ dB}} |G_{cl}(0)| \Leftrightarrow \frac{1}{\omega_{BW}^2 \tau^2 + 1} = \frac{1}{2} \Leftrightarrow \boxed{\omega_{BW} = \frac{1}{\tau}}$
- Rise time: $t_r \approx 2.2\tau$
- Setting time: $t_s \approx 4\tau$ Phase margin is always 90°

Control design based on specifications

Example: $G(s) = \frac{7.603}{s^2 + 11.35s + 25.59}$

Design a controller such that the closed-loop system has rise time $t_r \cong 0.11$ s, no overshoot and zero steady state error.

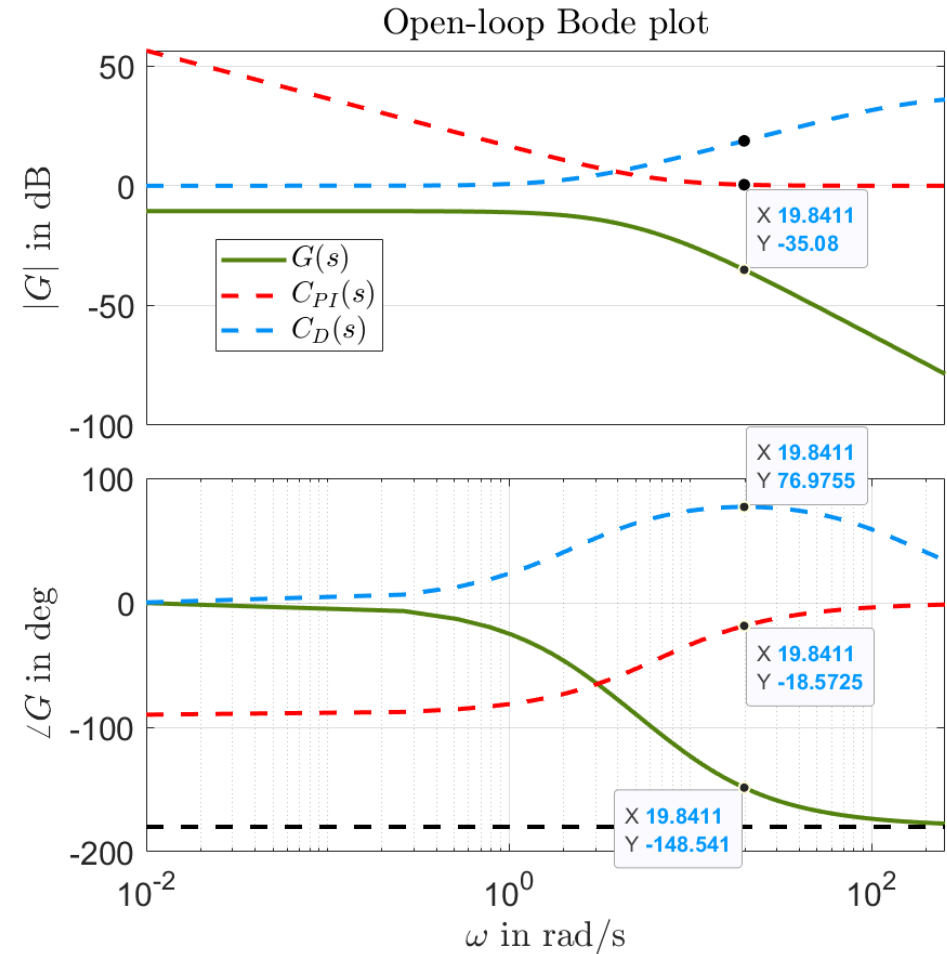
- All specifications indicate a design that leads to a first-order low-pass filter with $\tau = 0.05$ s.
- The new crossover frequency will be as the desired bandwidth: $\omega_c \approx 2.2/0.11 = 20$ rad/s .
- We will need an I-part such that we have zero steady state error.
- Phase margin is always $90^\circ \rightarrow$ We also need Lead.



Control design based on specifications

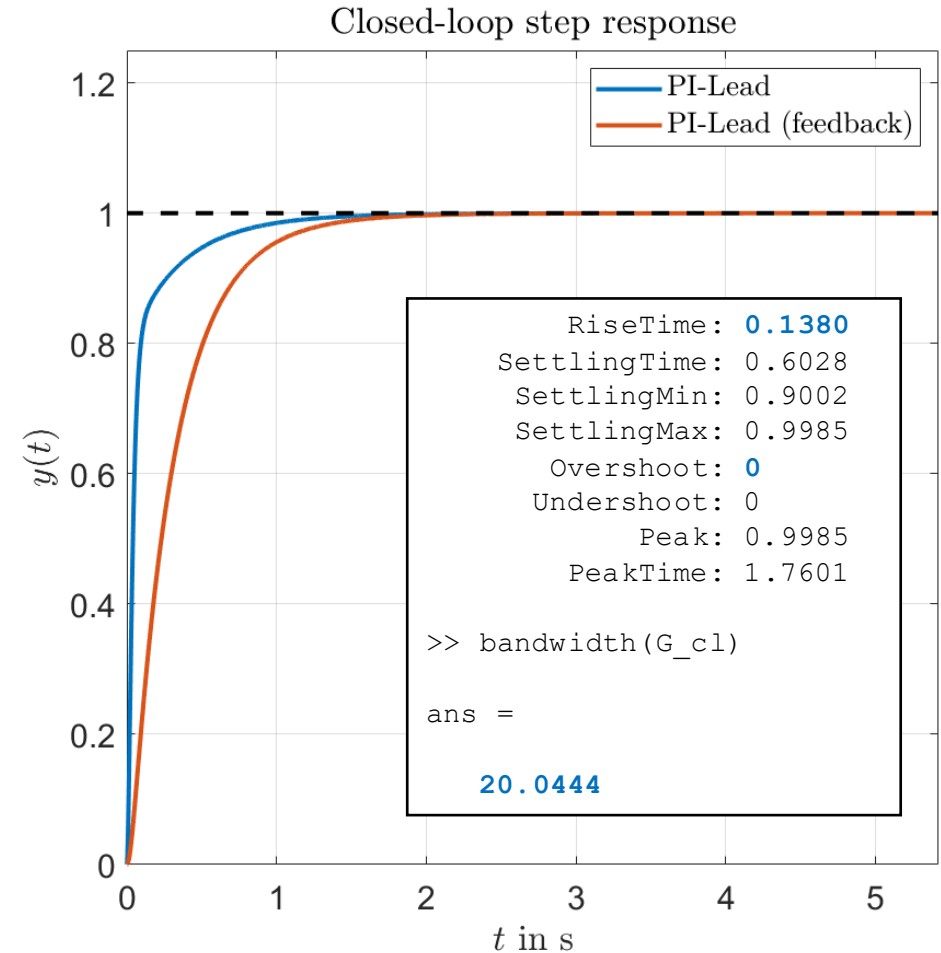
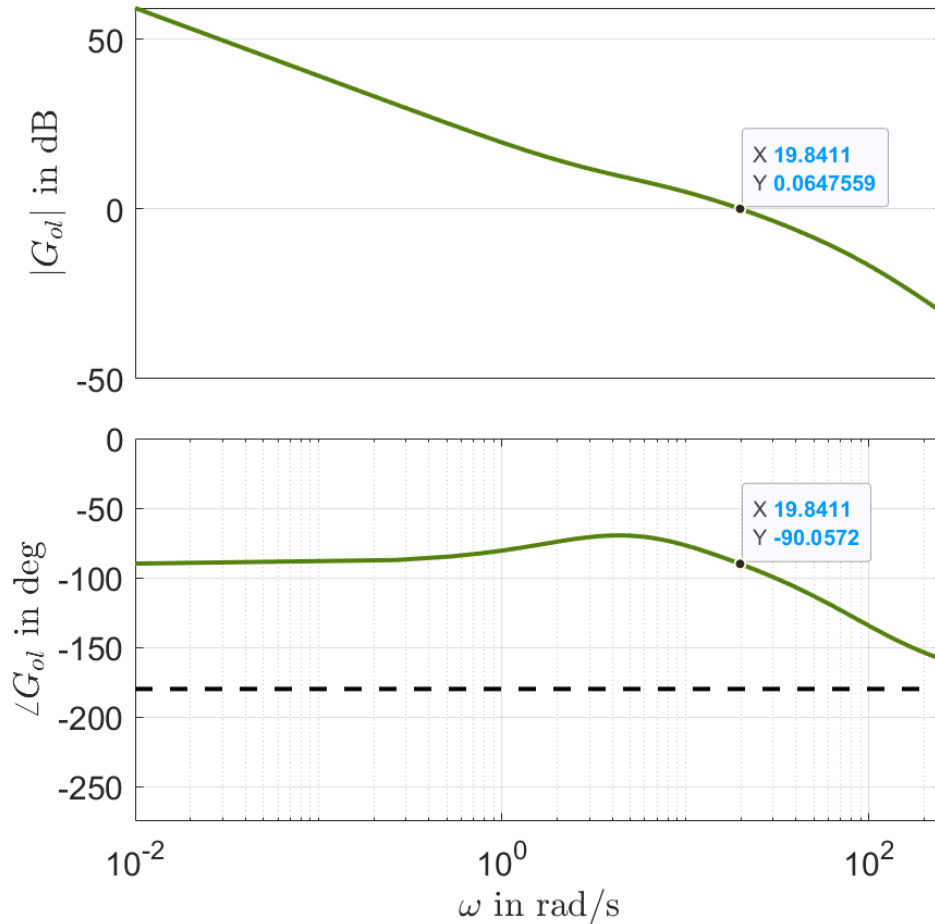
Example: $G(s) = \frac{7.603}{s^2 + 11.35s + 25.59}$

- Choose $N_i = 3 \Rightarrow \tau_i = \frac{N_i}{\omega_c} = 0.15$
- $-\pi + \frac{\pi}{2} = \angle G(20j) - \arctan\left(\frac{1}{3}\right) + \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) \Leftrightarrow$
 $\alpha = 0.013 \Rightarrow \tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.44$
- $K_P = \frac{1}{|C_{PI}(s)C_D(s)G(s)|} \Big|_{s=j\omega_c} = 6.24$



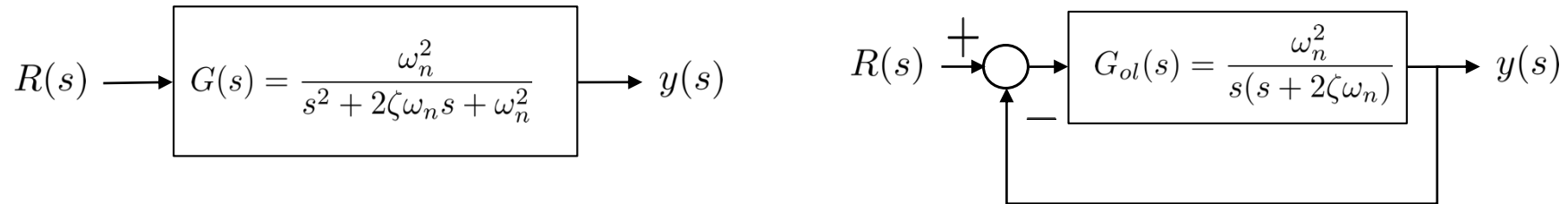
Control design based on specifications

Example_first_order_specifications_control.m



Control design based on specifications

- It is more realistic to require that a closed-loop system behave like a 2nd order lowpass filter with specified step response properties.
- For a second order system $G_{cl}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ we have:



$$\begin{aligned}
 t_r &\approx \frac{1.8}{\omega_n} \\
 t_p &= \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \\
 t_s &\approx \frac{4}{\zeta \omega_n} \\
 M_p &= e^{\frac{-\pi \zeta}{\sqrt{1 - \zeta^2}}}
 \end{aligned}$$

$$\begin{aligned}
 \omega_c &= \omega_n \sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2} \\
 \omega_{BW} &= \omega_n \sqrt{\sqrt{2 - 4\zeta + 4\zeta^4} + 1 - 2\zeta^2} \\
 \Rightarrow \omega_c &= \omega_{BW} \frac{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}}{\sqrt{\sqrt{2 - 4\zeta + 4\zeta^4} + 1 - 2\zeta^2}}
 \end{aligned}$$

Once we have ω_n and ζ , we get the open loop specifications we need for the design

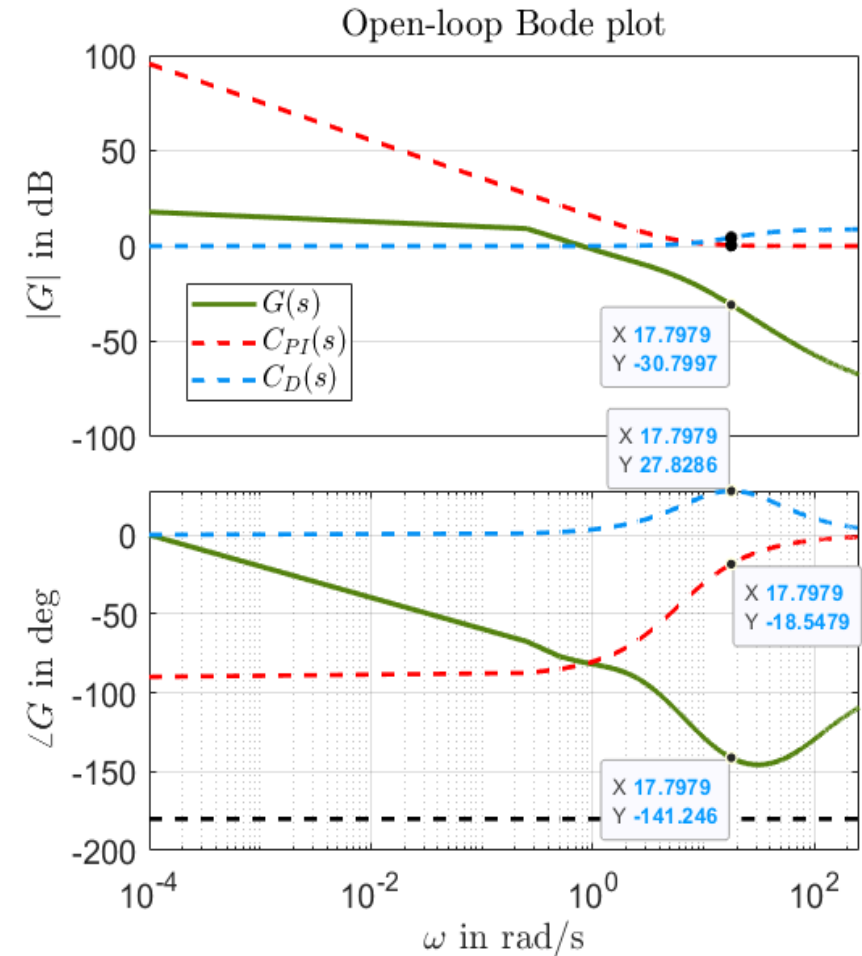
$$\Rightarrow \gamma_M = \frac{\pi}{2} - \arctan \left(\frac{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}}{2\zeta} \right) = \arctan \left(\frac{2\zeta}{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}} \right)$$

Control design based on specifications

Example: $G(s) = \frac{0.1(s+2)(s+100)}{(s+0.1)(s^2+11.35s+25.59)}$

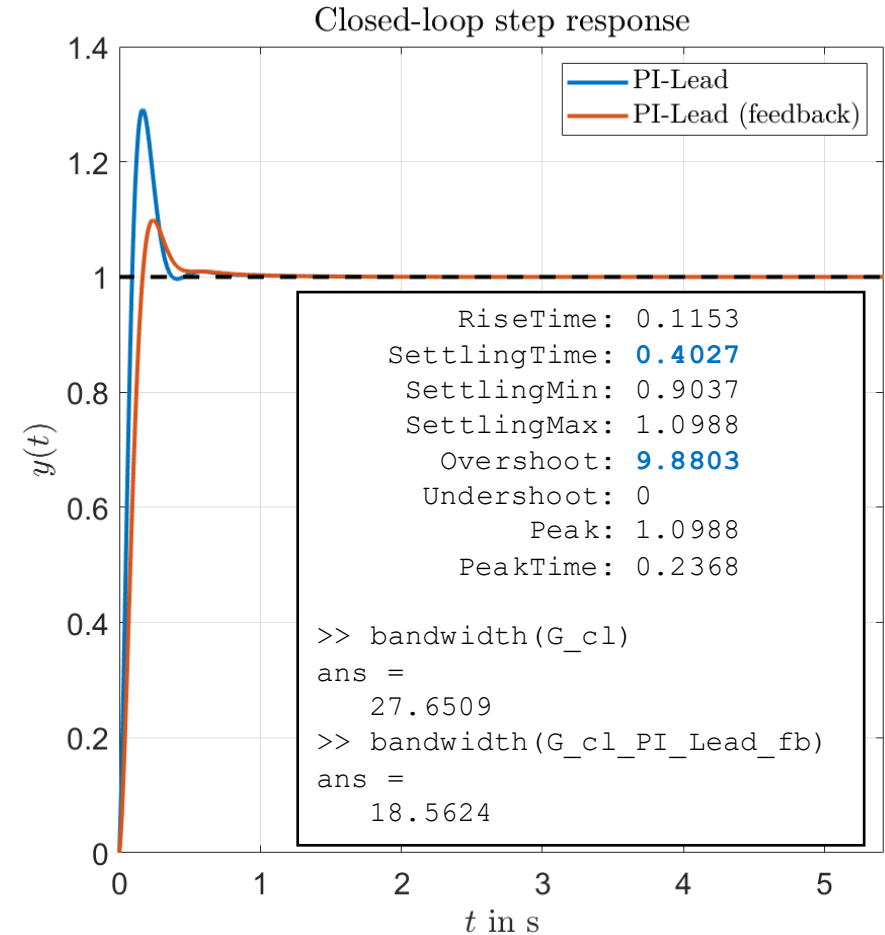
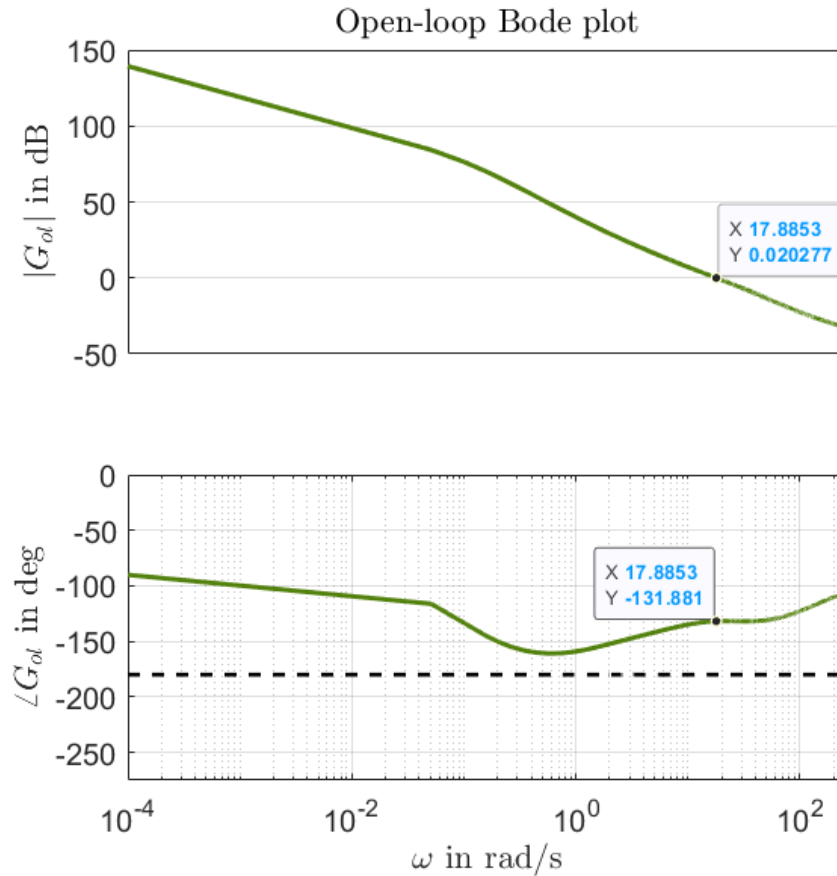
Design a controller such that the closed-loop system has settling time $t_s \cong 0.4$ s, overshoot $M_p \leq 20\%$ and zero steady state error.

- $M_p = 0.2 \Rightarrow \zeta = 0.456 \Rightarrow \gamma_M = 48.14^\circ$
- $t_s = 0.4 \Rightarrow \omega_n = 21.9 \Rightarrow \omega_c = 17.9 \text{ rad/s}$
- Choose $N_i = 3 \Rightarrow \tau_i = \frac{N_i}{\omega_c} = 0.1675$
- $-\pi + \gamma_M = \angle G(17.9j) - \arctan\left(\frac{1}{3}\right) + \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) \Leftrightarrow$
- $\alpha = 0.36 \Rightarrow \tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.092, K_P = \frac{1}{|G_{ol}(17.9j)|} = 20.1$



Control design based on specifications

Example_second_order_specifications_control.m



Control design based on specifications

		1 st order unit-gain lowpass filter	2 nd order unit-gain lowpass filter
Time domain specifications	Rise time	$t_r \approx 2.2\tau$	$t_r \approx \frac{1.8}{\omega_n}$
	Settling time	$t_s \approx 4\tau$	$t_s \approx \frac{4}{\zeta\omega_n}$
	Peak time	–	$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$
	Overshoot	–	$M_p = e^{\frac{-\pi\zeta}{\sqrt{1 - \zeta^2}}}$
Frequency domain specifications	Bandwidth	$\omega_{BW} = \frac{1}{\tau}$	$\omega_{BW} = \omega_n \sqrt{\sqrt{2 - 4\zeta + 4\zeta^4} + 1 - 2\zeta^2}$
	Phase margin	$\gamma_M = 90^\circ$	$\gamma_M = \frac{\pi}{2} - \arctan\left(\frac{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}}{2\zeta}\right) = \arctan\left(\frac{2\zeta}{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}}\right)$
	Crossover frequency	$\omega_c = \frac{1}{\tau}$	$\omega_c = \omega_n \sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}$

Control design based on specifications

What about the steady-state error?

- A system is called “*type-n*” if it has n integrators (and no zeros at the origin): $G(s) = \frac{1}{s^n} G_0(s)$
- If a P controller is used with unit feedback, the error transfer function is

$$G_e(s) = \frac{1}{1 + K_P G(s)} = \frac{s^n}{s^n + K_P G_0(s)}$$

No poles or zeros at the origin $\rightarrow G_0(0)$ is finite

- For a unit step input, the steady state error is

$$e_{ss} = \lim_{s \rightarrow 0} e(s) = \lim_{s \rightarrow 0} s \frac{s^n}{s^n + K_P G_0(s)} \frac{1}{s} = \lim_{s \rightarrow 0} \frac{s^n}{s^n + K_P G_0(s)} = \begin{cases} \frac{1}{1 + K_P G_0(0)} & , \text{ if } n = 0 \\ 0 & , \text{ if } n \geq 1 \end{cases}$$

Mini quiz: Why not just an integrator?

We need at least one integrator in the open loop $\rightarrow$ add a PI

- For a unit ramp input, the steady state error is

$$e_{ss} = \lim_{s \rightarrow 0} e(s) = \lim_{s \rightarrow 0} s \frac{s^n}{s^n + K_P G_0(s)} \frac{1}{s^2} = \lim_{s \rightarrow 0} \frac{s^{n-1}}{s^n + K_P G_0(s)} = \begin{cases} \infty & , \text{ if } n = 0 \\ \frac{1}{K_P G_0(0)} & , \text{ if } n = 1 \\ 0 & , \text{ if } n \geq 2 \end{cases}$$

Control design based on specifications

What about the steady-state error?

- If the system has n zeros at the origin (and no integrators): $G(s) = s^n G_0(s)$ → No poles or zeros at the origin → $G_0(0)$ is finite
- If a P controller is used with unit feedback, the error transfer function is

$$G_e(s) = \frac{1}{1 + K_P G(s)} = \frac{1}{1 + s^n K_P G_0(s)}$$

- For a unit step input, the steady state error is

$$e_{ss} = \lim_{s \rightarrow 0} e(s) = \lim_{s \rightarrow 0} s \frac{1}{1 + s^n K_P G_0(s)} \frac{1}{s} = \begin{cases} \frac{1}{1 + K_P G_0(0)} & , \text{ if } n = 0 \\ 1 & , \text{ if } n \geq 1 \end{cases}$$

We need at $n+1$ integrators in the open loop → add $n+1$ PI

- The open-loop transfer function will be

$$G_{ol}(s) = K_P \underbrace{\frac{\tau_{i,1}s + 1}{\tau_{i,1}s}}_{PI_1} \underbrace{\frac{\tau_{2,1}s + 1}{\tau_{2,1}s}}_{PI_2} \dots \underbrace{\frac{\tau_{i,n+1}s + 1}{\tau_{i,n+1}s}}_{PI_{n+1}} s^n G_0(s) = \frac{1}{s} K_P \underbrace{\frac{\tau_{i,1}s + 1}{\tau_{i,1}} \frac{\tau_{2,1}s + 1}{\tau_{2,1}} \dots \frac{\tau_{i,n+1}s + 1}{\tau_{i,n+1}}}_{G_{ol,0}} G_0(s) \triangleq \frac{1}{s} G_{ol,0}$$

All the PI zeros left of ω_c

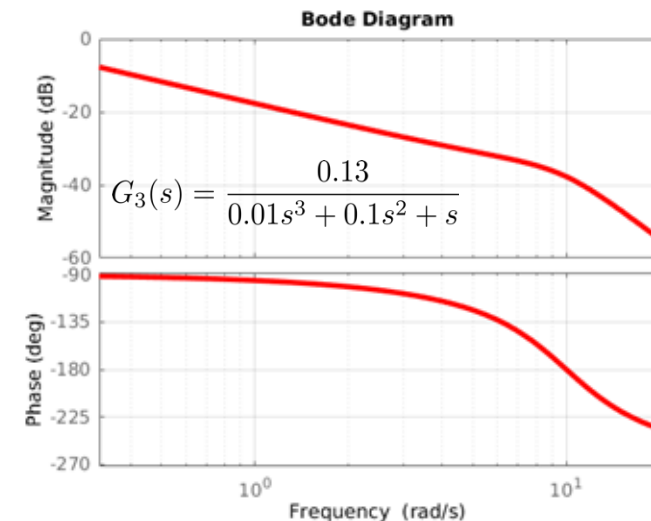
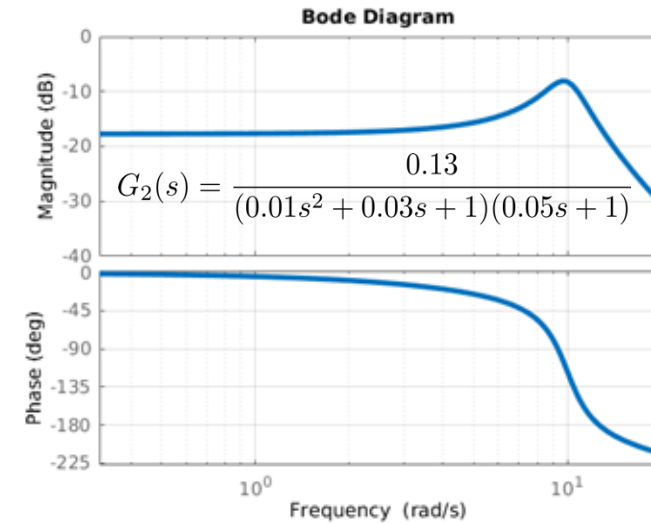
- The steady state error is $e_{ss} = \lim_{s \rightarrow 0} s \frac{1}{1 + G_{ol,0}(s)} \frac{1}{s} = 0$

Control design based on specifications

Quick assessment from open-loop Bode plot

- $G_2(s)$ has flat low frequency gain $\rightarrow$ type-0. We need a PI but we need to be careful with multiple crossover frequencies. The drop in the phase shift close to ω_π is steep $\rightarrow$ we need to use an “aggressive” (small α) Lead-part to improve the phase in that frequency range. We place the crossover frequency after the resonance peak location.
- $G_3(s)$ has large low frequency gain $\rightarrow$ type-1. We do not need a PI. The drop in the phase shift close to ω_π is mild $\rightarrow$ we can use Lead-part to improve the phase in that frequency range.

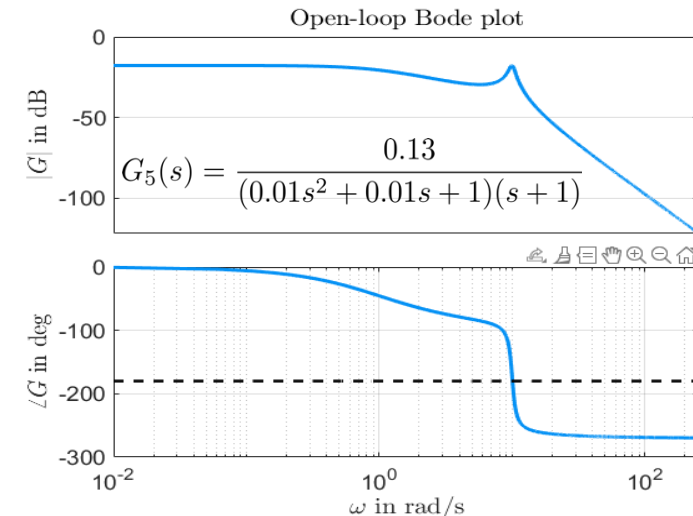
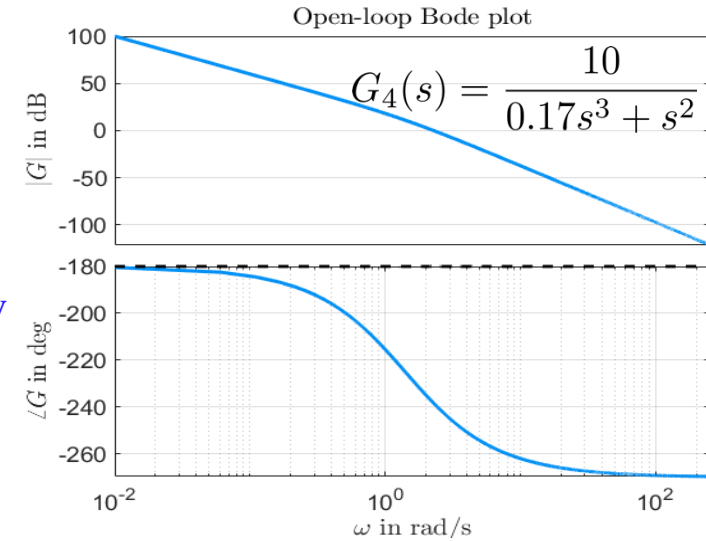
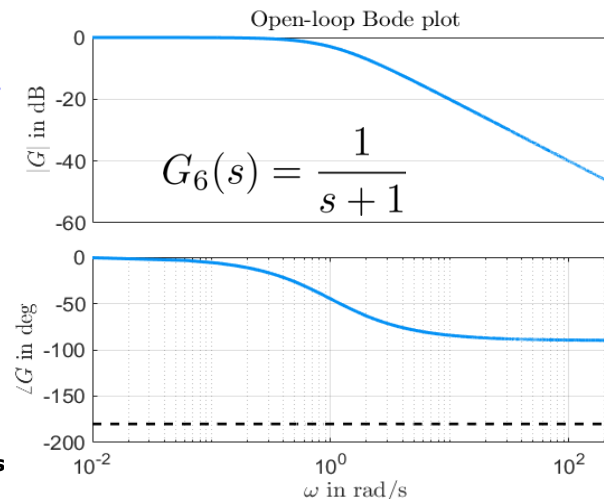
Mini quiz: How much phase can we add with a P-Lead?



Control design based on specifications

Quick questions

- Does $G_4(s)$ need an I-part to achieve zero static error with step inputs? → No, high gain at low frequencies
- If we use a P-Lead controller, will the closed-loop system be stable? → Yes, add phase around the crossover frequency
- For $G_5(s)$, is it a good idea to use a P-Lead to place the crossover frequency at the location of the resonance peak? → No really, too steep phase drop at ω_π .
- Does $G_6(s)$ (in theory) need a PI or a P-Lead? → No, we can arbitrarily increase K_P .



Control design based on specifications

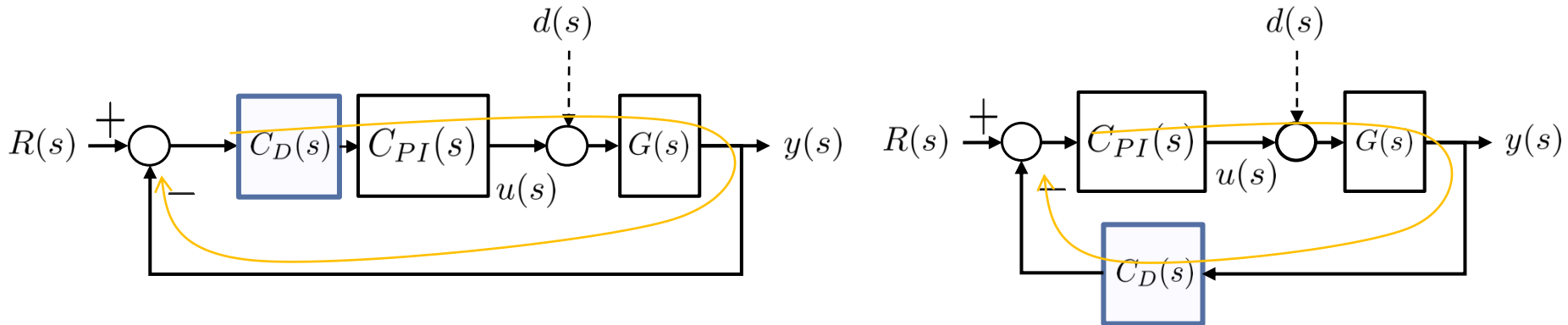
Comprehension questions

- Does a lower crossover frequency give a lower bandwidth?
 - In general, yes. Remember that $\omega_c \propto \omega_{BW}$
- Does a lower bandwidth give a longer settling time?
 - In general, yes. Remember that $t_s \cong 4/\zeta\omega_n$, $\omega_n \propto \omega_{BW}$
- Does a shorter settling time give a stronger control signal (u)?
 - In general, yes. Large (closed-loop) bandwidth $\rightarrow$ large (open-loop) crossover frequency $\rightarrow$ usually low magnitude of $G(s)$ $\rightarrow$ high K_P gain.
- Does a Bode plot for a closed-loop control system always start at 0 dB?
 - Not necessarily. When there is a non-zero static error, it does not start from 0 dB.
- Is the closed-loop bandwidth the same as the crossover frequency in open loop?
 - No, although they are usually close.

P-Lead in forward vs P-Lead in feedback

- Does the location of $C_D(s)$ matter for the open-loop calculations?
 - No, in both cases the open-loop is $C_{ol}(s) = C_D(s) C_{PI}(s) G(s)$.
- Does the location of $C_D(s)$ matter for the closed-loop performance?
 - Yes, the closed-loop bandwidth drops if $C_D(s)$ is placed to the feedback branch (why?).
- Does the location of $C_D(s)$ matter for the effect of an input disturbance $d(s)$ on the output?
 - No, in both architectures the transfer function from $d(s)$ to $y(s)$ is given by

$$G_{y,d}(s) = \frac{G(s)}{1 + C_D(s)C_{PI}(s)G(s)}$$



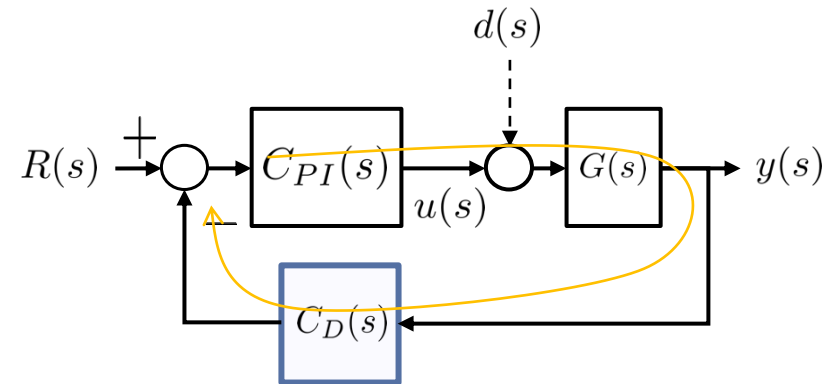
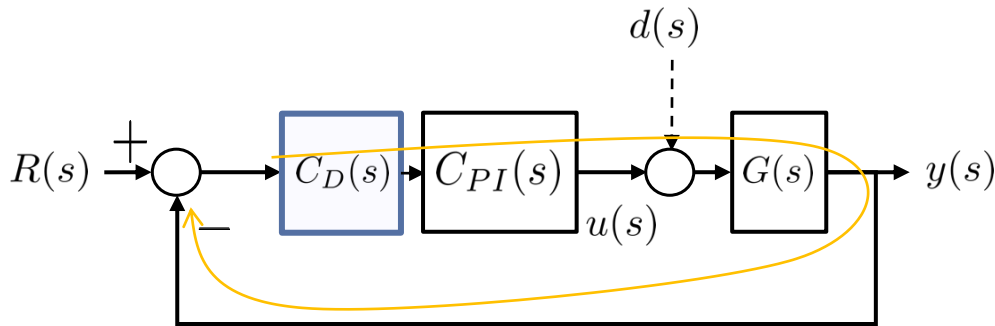
P-Lead in forward vs P-Lead in feedback

- Does the location of $C_D(s)$ matter for the control signal $u(s)$?
 - Yes, the transfer function from the reference $R(s)$ to the control signal $u(s)$ is given by:

$$G_{u,r}(s) = \frac{C_D(s)C_{PI}(s)}{1 + C_D(s)C_{PI}(s)G(s)} \quad \text{or}$$

$$G_{u,r}(s) = \frac{C_{PI}(s)}{1 + C_D(s)C_{PI}(s)G(s)}$$

$$C_D(s) = \frac{j\tau_d\omega + 1}{j\alpha\tau_d\omega + 1}, \quad \alpha < 1 \quad \rightarrow \text{Large amplitude for fast reference } R(s) \text{ (large } \omega)$$



Some extra considerations

- If we choose not to include a controller part, we do not include it into the phase calculation, e.g.
 $-180^\circ + \gamma_M = \phi_G + \phi_m$ (no PI) or $-180^\circ + \gamma_M = \phi_G + \phi_i$ (no Lead)
- For type-0 open-loop systems (with no PI in the loop), smaller static error implies lower phase margin.
- When during the controller design the contribution from the Lead part is larger than 90° , we can introduce additional Lead terms to complement the extra needed positive phase, e.g.

$$\phi_m = -180^\circ + \gamma_M - \phi_G = 130^\circ$$

$$\phi_m = \phi_{m,1} + \phi_{m,2} = 70^\circ + 60^\circ$$

$$\begin{cases} \phi_{m,1} = 70^\circ \Rightarrow \alpha_1 = \dots \Rightarrow \tau_{d,1} = \dots C_{D,1} = \dots \\ \phi_{m,2} = 40^\circ \Rightarrow \alpha_2 = \dots \Rightarrow \tau_{d,2} = \dots C_{D,2} = \dots \end{cases} \Rightarrow C_D(s) = C_{D,1}(s)C_{D,2}(s)$$

- Very frequently we use the two following terms
 - Loop gain (static loop gain): $|G_{ol}(j\omega)|$ (or for static $|G_{ol}(0)|$)
 - Closed-loop gain (static closed-loop gain): $|G_{cl}(j\omega)|$ (or for static $|G_{cl}(0)|$)

Today you learned

- How to transform closed-loop specifications into open-loop specifications for design.
- What bandwidth is and how it affects the closed-loop system's performance.
- What “type – n” systems are and how pure integrators/differentiators affect the steady-state error.
- How to interpret Bode plots for control design.
- How the placing of the Lead part affects the bandwidth and why.

Next lecture 10

Unstable systems.

- Explain the concept of instability in dynamical systems
- Use Nyquist plots to stabilise open-loop unstable systems

Introduction to mandatory assignment

- Basics of REGBOT balance
- Introduction to cascaded control for balancing motion
- Guidelines and tips for the assignment

Today's Exercises

Group work: Continue REGBOT position control from Day 8

Home alone: Theoretical exercises LCD1: 1, 2, 4, 5, 6, 7, 9